



UNIVERSITÀ
DI TRENTO

What are the DOM and the BOM?

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From a page to its BOM-DOM

history

navigator

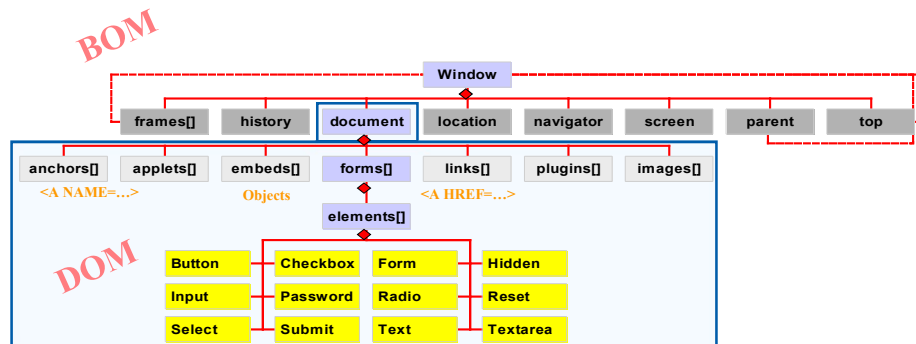
location

document

window

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Hierarchy



Symbol ◆ means containment (has-a)

Dashed line means "is an instance of"

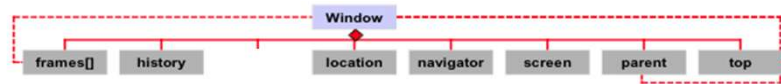
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What is the BOM, and which are its components?

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BOM

Window

*"An open web browser window or frame"*

Main properties

history
frames[]
document
location
navigator
screen
parent – top

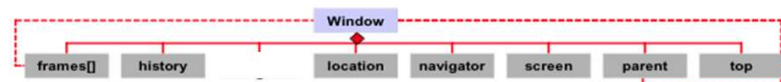
Other properties

status – defaultStatus
name

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BOM

Window



Main methods

alert(), prompt(), confirm()
focus(), blur()
moveBy(), moveTo()
resizeBy(), resizeTo()
scrollBy(), scrollTo()
setInterval(), clearInterval()
setTimeout(), clearTimeout()

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Some Examples

https://www.w3schools.com/jsref/tryit.asp?filename=tryjsref_alert

https://www.w3schools.com/jsref/tryit.asp?filename=tryjsref_win_moveto

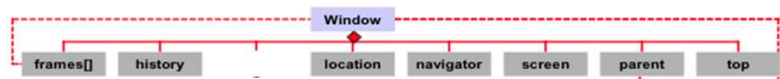
https://www.w3schools.com/jsref/met_win_setinterval.asp

https://www.w3schools.com/jsref/met_win_settimeout.asp

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Navigator

BOM



"Information about the browser in use"

Main properties

appName

appVersion

cookieEnabled

platform

Main methods

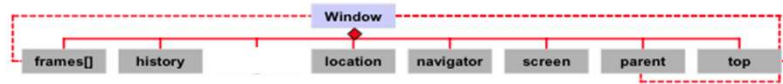
javaEnabled()

https://www.w3schools.com/jsref/tryit.asp?filename=tryjsref_nav_language

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BOM

History

*"The URL history of the browser"*

Main properties

length

Main methods

back()

forward()

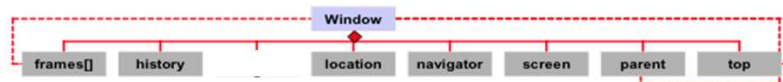
go(+/-n)

go(target_substring)

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BOM

Location

*"The specification of the current URL"*

Main properties

href

protocol, hostname, port

search

hash

Main methods

reload()

replace()

https://www.w3schools.com/jsref/tryit.asp?filename=tryjsref_loc_replace

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What is the DOM?

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What is the DOM?

- When a webpage is loaded, the browser creates a Document Object Model (DOM) of the page itself
- The DOM is a W3C (World Wide Web Consortium) standard for accessing documents:
“The W3C Document Object Model (DOM) is a platform and language-neutral interface that allows programs and scripts to dynamically access and update the content, structure, and style of a document”
- The W3C DOM standard is separated into 3 different parts, we will focus on:
HTML DOM - standard model for HTML documents

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What is the DOM?

- The HTML DOM is a standard object model (is an object-oriented representation of the structure and content of a web page) and programming interface for HTML.
- It defines:
 - The HTML elements as objects
 - The properties of all HTML elements
 - The methods to access all HTML elements
 - The events for all HTML elements

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JS and the DOM

Every element in a document—the document as a whole, the head, tables within the document, table headers, text within the table cells—is part of its DOM, so they can all be accessed and manipulated using the DOM and a scripting language like JavaScript

With the object model, JavaScript can:

- change all the HTML (elements, attributes, styles) in the page
- add or remove existing HTML elements and attributes
- react to HTML events in the page
- create new HTML events in the page

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Using languages

- Implementations of the DOM can be built for any language:
e.g. Javascript, Java, Python and so on
- But Javascript is the only one that can work client-side

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**What are the fundamental
components in DOM?**

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Fundamental datatypes - 1

- **Node** (Abstract base class)
Every object located within a document is a node. In an HTML document, an object can be a **document**, an **element node** but also a **text node** or **attribute node**
- **Document** (is-a Node)
The root document object itself. Document contains a main child node (an element node), which is the <html> element.
From there, the entire hierarchy develops.
- **Element** (is-a Node)
The most general base class from which all element objects (i.e. objects that represent elements) in a Document inherit

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Fundamental datatypes - 2

- **Attr** (is-a Node)
Attr object represents an HTML attribute
- **NodeList** (has Node(s))
A nodeList is an array of nodes. Items in a nodeList are accessed by index: Node_list_name.item(1) or Node_list_name[1]
- **NamedNodeMap**
A namedNodeMap is an associative array, where the items are accessed by name. They can also be accessed by index using the item() method (but nodes are in no particular order in the list). You can also add and remove items from a namedNodeMap.

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About Nodes yet

- All of the following types inherit the Node interface's methods and properties:

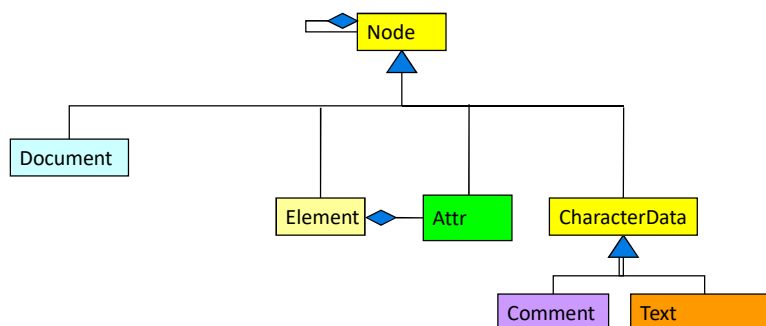
Document, **Element**, **Attr**, **CharacterData** (which **Text**, **Comment**, and **CDATASection** inherit), **ProcessingInstruction**, **DocumentFragment**, **DocumentType**, **Notation**

Text (is-a Node): a text object represents the text inside a HTML element. A text node contains only a string. It may not have children and is always a leaf of the tree

see <https://developer.mozilla.org/en-US/docs/Web/API/Node>

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The Node hierarchy



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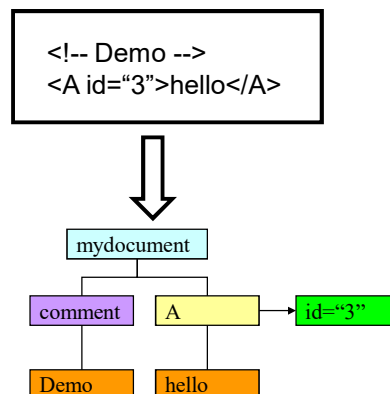
Node: WARNING!

The implied semantic of this model is WRONG!

You might deduce that a comment might contain another comment, or a document, or any other node!

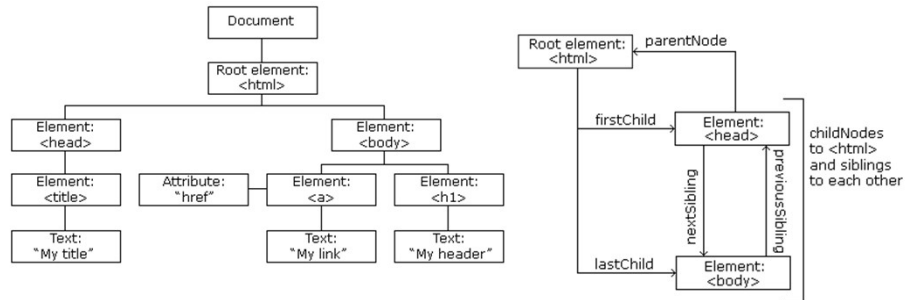
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Example of a tree



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Another example



source: https://www.w3schools.com/js/js_htmlDOM_navigation.asp

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Node properties

- **Node.nodeType** - Read only (see Table)
- **Node.nodeName** - Read only
(An HTML element will contain the name of the corresponding tag, like 'audio' for an HTMLAudioElement, a Text node will have the '#text' string, or a Document node will have the '#document' string).
- **Node.baseURI** - Read only
- **Node.textContent** - Read/Write

Node Type	
ELEMENT_NODE	1
ATTRIBUTE_NODE	2
TEXT_NODE	3
CDATA_SECTION_NODE	4
ENTITY_REFERENCE_NODE	5
ENTITY_NODE	6
PROCESSING_INSTRUCTION_NODE	7
COMMENT_NODE	8
DOCUMENT_NODE	9
DOCUMENT_TYPE_NODE	10
DOCUMENT_FRAGMENT_NODE	11
NOTATION_NODE	12

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https://www.w3schools.com/jsref/tryit.asp?filename=tryjsref_node_nodetype

Node: Read only navigation properties

- Node.childNodes
- Node.firstChild
- Node.lastChild

- Node.nextSibling
- Node.previousSibling

- Node.parentNode
- Node.parentElement

https://www.w3schools.com/jsref/tryit.asp?filename=tryjsref_node_childnodes

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Other examples

https://www.w3schools.com/js/tryit.asp?filename=tryjs_dom_nav_innerhtml2

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